

Motivation : CLT

Def. Standard Normal distribution: A continuous r.v Z is said to have the standard Normal distribution if its PDF ϕ is given by

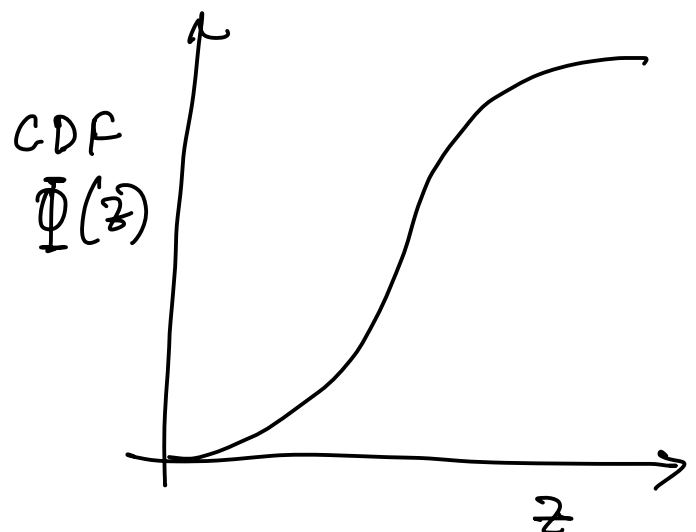
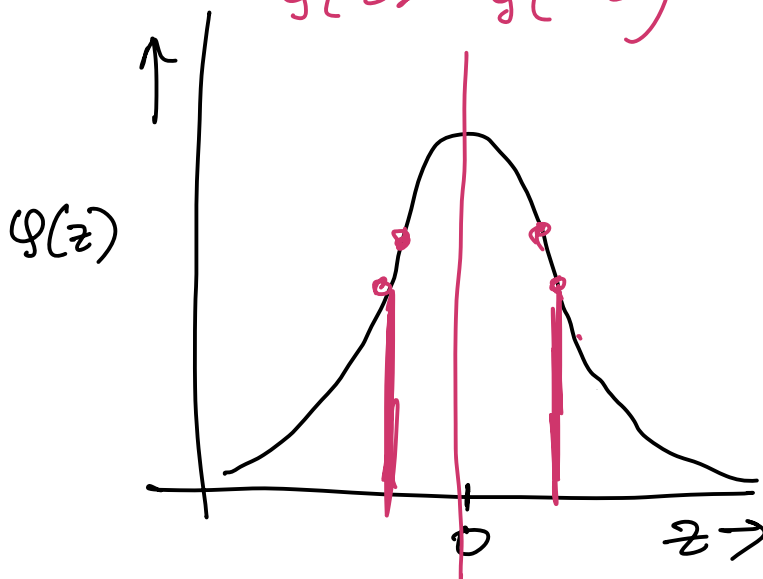
$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

$$Z \sim N(0,1)$$

The standard normal CDF Φ is given by

$$P(Z \leq z) = \Phi(z) = \int_{-\infty}^z \phi(t) dt = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

$$\phi(z) = \phi(-z)$$



Deriving the normalization constant $\sqrt{2\pi}$

Valid PDF: $\int_{-\infty}^{\infty} \phi(t) dt \stackrel{?}{=} 1$

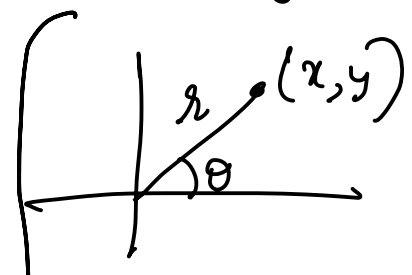
$$\int_{-\infty}^{\infty} \left(\frac{1}{\sqrt{2\pi}} \right) e^{-t^2/2} dt \stackrel{?}{=} 1$$

$$I = \int_{-\infty}^{\infty} e^{-t^2/2} dt = \sqrt{2\pi}$$

$$I \cdot I = I^2 = \int_{-\infty}^{\infty} e^{-x^2/2} dx \int_{-\infty}^{\infty} e^{-y^2/2} dy$$
$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-x^2/2} e^{-y^2/2} dx dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2+y^2)/2} dx dy$$

$$(x, y) \longleftrightarrow (r, \theta)$$



$$r^2 = x^2 + y^2$$

$$= \int_0^{\infty} \int_0^{2\pi} e^{-r^2/2} \cdot r \, dr \, d\theta$$

Check this change of variable

$$\frac{r^2}{2} = t$$

$$= \int_0^{\infty} e^{-t} \, dt \int_0^{2\pi} d\theta$$

$$= 2\pi$$

$$I = \sqrt{2\pi}$$

Properties of standard normal distribution

$$(i) \quad \phi(z) = \phi(-z)$$

$$(ii) \quad \Phi(z) = 1 - \Phi(-z)$$

$$(iii) \quad \text{If } z \sim N(0,1), \text{ then } -z \sim N(0,1)$$

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$

Deriving the mean & variance

$$E[z] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z e^{-z^2/2} dz = 0$$

$$V(z) = \underbrace{E[z^2]} - \cancel{(E[z])^2} \rightarrow 0$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} z^2 e^{-z^2/2} dz$$

$$= \frac{2}{\sqrt{2\pi}} \int_0^{\infty} z^2 e^{-z^2/2} dz$$

$$= \frac{2}{\sqrt{2\pi}} \cdot \frac{\sqrt{2\pi}}{2} = 1$$

General Normal distribution (μ, σ)

Defⁿ: (Normal distribution) If $Z \sim N(0, 1)$, then $X = \mu + \sigma Z$ is said to have the Normal distribution with mean μ and variance σ^2 . We denote this by $X \sim N(\mu, \sigma^2)$.

Vice versa, if $X \sim N(\mu, \sigma^2)$, then $\frac{X - \mu}{\sigma} \sim N(0, 1)$

Thm: Normal (CDF & PDF). let $X \sim N(\mu, \sigma^2)$
Then the CDF of X ?
$$F(x) = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

and the PDF of X is

$$f(x) = \phi\left(\frac{x - \mu}{\sigma}\right) \cdot \frac{1}{\sigma}$$

Pf: $F(x) = P(X \leq x)$

$$= P\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right)$$

$$Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

$$= \Phi$$

$$P\left(Z \leq \frac{x - \mu}{\sigma}\right)$$

$$= \Phi\left(\frac{x - \mu}{\sigma}\right)$$

$$F(x) = P(X \leq x)$$

$$P(X \leq x+2) = F(x+2)$$

$$CDF = \Phi\left(\frac{x - \mu}{\sigma}\right)$$

$$PDF = (CDF)' = \frac{d}{dx} \Phi\left(\frac{x - \mu}{\sigma}\right)$$

$$= \phi\left(\frac{x - \mu}{\sigma}\right) \cdot \frac{1}{\sigma}$$

Ex. Let $X \sim N(-1, 4)$

$$P(|X| < 3)$$

$$Z = \frac{X + 1}{2} \sim N(0, 1)$$

$$P(Z < x) = \Phi(x)$$

$$= P(-3 < \cancel{X} < 3)$$

$$= P\left(\frac{-3+1}{2} < \frac{X+1}{2} < \frac{3+1}{2}\right)$$

$$\approx P\left(-1 < \frac{X+1}{2} < 2\right)$$

$$= P(-1 < Z < 2)$$

$$= \Phi(2) - \underbrace{\Phi(-1)}$$

$$\Phi(z) = 1 - \Phi(-z)$$

$$P(|X| < 3) = \underbrace{\Phi(2)} + \underbrace{\Phi(1)} - 1$$

Exponential distribution

Continuous counterpart to the *geometric distribution*.

Models the time between independent events occurring at a constant average rate.

Examples:

1. Time until a radioactive particle decays or a light bulb burns out.
2. Time between customers arriving at a service desk
3. Time between successive mutations on a DNA strand

Defⁿ: A continuous random variable X is said to have an exponential distribution with rate parameter λ ($\lambda > 0$) if

PDF: $f(x) = \begin{cases} \lambda e^{-\lambda x} & (x \geq 0) \\ 0 & ; x < 0 \end{cases}$

Check: Valid PDF $\int_0^{\infty} f(x) dx = 1 \quad \checkmark$

CDF: $F(x) = P(X \leq x) = \int_0^x f(t) dt \quad x \geq 0$

$$= \int_0^x \lambda e^{-\lambda t} dt \quad x \geq 0$$

$$= 1 - e^{-\lambda x} \quad ; x \geq 0$$

Standardizing exponential

Let $Y = \lambda X$, then $X \sim \text{Expo}(\lambda)$

show $Y \sim \text{Expo}(1)$

Pf: $P(Y \leq y) = P(\lambda X \leq y)$

$$= P\left(X \leq \frac{y}{\lambda}\right)$$

$$= 1 - e^{-\lambda \cdot \frac{y}{\lambda}} = 1 - e^{-y}$$

$\Rightarrow Y \sim \text{Expo}(1)$

let $Y \sim \text{Expo}(1)$, find

$$E[Y] = \int_0^{\infty} y e^{-y} dy = 1$$

$$\begin{aligned} \text{Var}(Y) &= E[Y^2] - (E[Y])^2 \\ &= \int_0^{\infty} y^2 e^{-y} dy \xrightarrow{\quad} 1 = 1 \end{aligned}$$

So, $X = \frac{Y}{\lambda}$ has $E[X] = \frac{1}{\lambda} E[Y] = \frac{1}{\lambda} \cdot 1 = 1/\lambda$

$\text{Var}(X) =$

$$\text{Var}\left(\frac{Y}{\lambda}\right) = \frac{1}{\lambda^2} \text{Var}(Y) = \frac{1}{\lambda^2}$$

Memoryless property

The most unique feature of the exponential distribution is that it is memoryless.

No matter how long you have waited, it is like starting fresh.

If there is a random variable (waiting time) for something to happen

like a phone call
already waited for 10 minutes,

the probability that it arrives in the next 5 minutes = the probability that it would arrive in the first 5 minutes of your wait.

The "object" does not age / remember how much time has passed.

Mathematically,

$$P(X \geq s+t \mid X \geq s) = P(X \geq t) \quad \text{memoryless property}$$

Here, $P(X \geq s) \rightarrow$ also called survival function.

Exponential satisfies the memoryless property

Pf: $LHS = P(X \geq s+t \mid X \geq s)$

$$= \frac{P(X \geq s+t, X \geq s)}{P(X \geq s)}$$

$$= \frac{P(X \geq s+t)}{P(X \geq s)} = \frac{1 - P(X \leq s+t)}{1 - P(X \leq s)}$$

$$= \frac{1 - (1 - e^{-\lambda(s+t)})}{1 - (1 - e^{-\lambda s})} = \frac{e^{-\lambda s} e^{-\lambda t}}{e^{-\lambda s}} = e^{-\lambda t} = P(X \geq t)$$

Problems :

= RHS.

- ① Customers arrive at a bank at an average rate of 6 per hour. What is the probability that the next customer arrives within 15 minutes?

Solution : $\lambda = 6$

$$P(X \leq 0.25)$$

$$= 1 - e^{-6 \cdot 0.25}$$
$$= 1 - e^{-3/2} \quad \underline{\text{Ans}}$$

- ② You enter a post office to mail a package. There are two clerks, and both are currently busy serving two other customers

Clerk 1 Clerk 2
↓ ↓
Cust A Cust B

You are first in line & will be served as soon as either Cust A or Cust B leaves.

Suppose the service time for any customer follows

an Exponential distribution with a mean of 10 minutes.

Question:

When you sit down to wait, you notice that Cust A has already been at the counter for 20 minutes,

Cust B just arrived 1 min before you.

① Find the probability that Cust A finishes before Cust B? $= 0.5$

② What is your expected waiting time until you get to a clerk?

$$T = \min(X_A, X_B)$$

Figure out distribution of T ?

$X_1, \dots, X_n$ are i.i.d s.v.s with

$\text{Expo}(\lambda_1), \dots, \text{Expo}(\lambda_n)$

$$T \sim \text{Expo}\left(\frac{1}{10} + \frac{1}{10}\right)$$

$$\Rightarrow T \sim \text{Expo}\left(\frac{1}{5}\right)$$

$$E[T] = 5 \text{ mins}$$

✓

$\xrightarrow{\quad} \text{Exp}(\lambda_1), \dots, \text{Exp}(\lambda_n)$

$X_1, \dots, X_n \rightarrow \text{rates } \lambda_1, \dots, \lambda_n$

Claim

$$\min(X_1, \dots, X_n) \sim \text{Exp}(\lambda_1 + \dots + \lambda_n)$$

$$Y = \min(X_1, \dots, X_n)$$

$$P(Y \geq y) = P(\min(X_1, \dots, X_n) \geq y)$$

$$= P(X_1 \geq y, X_2 \geq y, \dots, X_n \geq y)$$

independ.

$$= P(X_1 \geq y) P(X_2 \geq y) \dots P(X_n \geq y)$$

$$= e^{-\lambda_1 y} \cdot e^{-\lambda_2 y} \dots e^{-\lambda_n y}$$

$$= e^{-(\lambda_1 + \dots + \lambda_n)y}$$

$$\Rightarrow Y \sim \text{Exp}(\lambda_1 + \dots + \lambda_n)$$

③ A high-traffic web server relies on 2 parallel cooling fans. The time until the first fan fails follows an exponential distribution because it depends on the minimum of 3 independent internal components, each failing at a rate of $\lambda = \underline{0.02}$ failures per month.

① Find the probability that the cooling system operates without any component failure for at least 20 months

$$\lambda = 0.02 + 0.02 + 0.02 = 0.06$$

$$P(X \geq 20) = e^{-\lambda \cdot 20} \\ = e^{-0.06 \cdot 20} = 0.301$$

$N(5, 1)$



② What is the probability that the technician takes between 4 and 7 hours to arrive?

$$\begin{aligned}
 P(4 < T < 7) &= \\
 &= P\left(\frac{4-5}{1} < \frac{T-5}{1} < \frac{7-5}{1}\right) \\
 Z \sim N(0,1) &\quad \underline{\hspace{10em}} \\
 &= P(-1 < Z < 2) \\
 &= \Phi(2) - \Phi(-1)
 \end{aligned}$$

③ If the server can only run safely for 6 hours after a fan failure starts, what is the probability that the technician arrives in time to prevent a total server meltdown.

$$\begin{aligned}
 P(T \leq 6) &= P\left(\frac{T-5}{1} \leq \frac{6-5}{1}\right) \\
 Z \sim N(0,1) &\quad \underline{\hspace{10em}} \\
 &= P(Z \leq 1) \\
 &= \Phi(1)
 \end{aligned}$$

